8.3.3 Demystifying document embeddings

TF–IDF is a simple form that tries to capture semantic meaning in documents. However, it's unreliable because it only counts word frequency and doesn't understand the relationships between words. A better and more modern method uses document embedding, a form of document vectorizing that better preserves the semantic meaning of the document.

Embedding networks are constructed by training neural networks on large datasets to map words, sentences, or documents to high-dimensional vectors, capturing semantic and syntactic relationships based on context and relationships in the data. You typically use a pretrained model trained on massive datasets to embed documents and perform embeddings. Models are available from many sources, including Hugging Face and, of course, OpenAI.

In our next scenario, we'll use an OpenAI embedding model. These models are typically perfect for capturing the semantic context of embedded documents. Listing 8.4 shows the relevant code that uses OpenAI to embed the documents into vectors that are then reduced to three dimensions and rendered into a plot.